QUALITY ASSURANCE & COMPREHENSIVE INSPECTION PLAN (QAP/QC)

BI-PARTING SLIDING HANGER DOOR

Company Name:	NIHVA Technologies Pvt. Ltd. Pune	**Vender Name:**	Royal Enterprises
Project Name:	Bi-Parting Sliding Hanger Door	**Drawing No:**	DWG NO-NTPL-0007-PM-001
Project No / Rev:	NTPL-0007	**Door Qty & Location:**	02 Leaf Sets Each door, Door qty - 02| Site: Hyderabad
Steel Grade / Spec:	YST 250 MPa (IS 1161 / IS 4923 / IS 2062 E250)	**Approved Brands:**	TATA Steel, SAIL, VIZAG, JINDAL (JSW)
Welding Process:	CO2 Welding ONLY (MIG/MAG - AWS A5.18 ER70S-6)	**Surface & Coating:**	2 Coats Zinc Chromate Epoxy Primer + 2 Coats PU Topcoat RAL - 7035
Overall Dimensions:	Width 8.6M x Height 14M Width 7.3M x Height 14M	**Primary Standards:**	IS 800, IS 1852, IS 2062, AWS D1.1

1. Scope & Approved Raw Material Standards

This Quality Assurance Plan (QAP) enforces manufacturing and inspection criteria for the fabrication and assembly of the W 8.6M x H 14M & W 7.3 M x H 14M Bi-Parting Sliding Hanger Door (DWG NO-NTPL-0007-PM-001). All structural members (Channels, Angles, Plates, Hollow Sections) strictly conform to Grade YST 250 MPa (Min Yield Strength: 250 N/mm², Tensile Strength: 410-540 N/mm²), procured from approved primary steel manufacturers: TATA Steel, SAIL, VIZAG (RINL), or JINDAL (JSW Steel) With all respective MTC. ALL structural joints must be executed exclusively using CO2 Welding (MIG/MAG process).

2. Engineering Profile & Technical Specifications Summary

Component / Member	Drawing Size & Specification	Approved Makes & Grade	Key Quality Inspection Points
Main Columns / Vertical Beams	ISMC 125 x 65 mm Channels Cutting: 6000mm (x2), 2000mm (x2), 4500/3500mm	YST 250 MPa / E250 TATA / SAIL / VIZAG / JINDAL	Web depth, flange width, squareness, zero sweep/camber, 100% mill test certificate match
Primary Structural Diagonal Truss	ISA 100 x 100 x 6 mm Angle Cutting Length: 1200mm grid layout	YST 250 MPa / E250 TATA / SAIL / VIZAG / JINDAL	Mitre cut bevel accuracy, root fit-up gap (1.5-2.0 mm), CO2 welding throat thickness
Secondary Lattice Bracing	ISA 50 x 50 x 4 mm Angle Top framing & diagonal lattice	YST 250 MPa / E250 TATA / SAIL / VIZAG / JINDAL	Angle profile straightness, CO2 continuous weld compliance, removal of sharp edges
Outer Connection Welding Plates	270 x 100 x 8 mm Thk Outer Plates Gap spacing: 500.00 mm C/C	YST 250 MPa / E250 TATA / SAIL / VIZAG / JINDAL	Thickness verification (8.0 mm ± 0.3 mm), pitch spacing (500 mm ± 1 mm), flush CO2 perimeter fillet weld
Motor Bracket Mounting Plate	275 x 200 x 16 mm Thk Plate Detail C (1:16) / Detail D (1:15)	YST 250 MPa / E250 TATA / SAIL / VIZAG / JINDAL	100% Dye Penetrant testing on heavy CO2 welds, face flatness, mounting hole layout
Hole Drilling Specifications	Ø 14.00 mm All Holes (+0.50 / -0.50 mm) Position: ⌖ Ø 0.5 mm @ MMC (Datum A)	Standard Machining (Drill Jigs Mandatory)	Go/No-Go plug gauge check (13.50-14.50 mm), true position relative to Datum A, 100% deburring

Wicket Access Door (Leaf 02)	Size: 1324 x 2060 mm Pass Door Frame Profiles	YST 250 MPa / E250 TATA / SAIL / VIZAG / JINDAL	CRITICAL NOTE: Strictly DO NOT weld in shop; cut to size and weld on side profiles at site using CO2 welding

3. Detailed Stage-Wise Quality Inspection Plan (QAP Control Matrix)

Stage	Inspection Checkpoint	Quality Standard / Requirement	Tolerance / Limit	Verification Method	QC Document Record
1. Raw Material	Manufacturer Brand Verification	Steel from TATA Steel, SAIL, VIZAG (RINL), or JINDAL (JSW)	100% Original Mill Stamp / Stencil Verification	Visual Inspection & Heat No. Verification	Material Receiving Report (MRR)
1. Raw Material	Material Grade & Mill Test Certificate	YST 250 MPa (Yield ≥ 250 N/mm², Tensile 410-540 N/mm², Elongation ≥ 23%)	100% Batch Match vs MTC Heat Numbers	MTC Review & Chemical/Mechanical Analysis	MTC Clearance Certificate
1. Raw Material	Dimensional & Surface Tolerances	ISMC 125x65, ISA 100x100x6, ISA 50x50x4, Plates 8mm/16mm	As per IS 1852 / IS 1161. Thickness: ± 0.3 mm	Vernier Caliper, Micrometer, Ultrasonic Thickness	Raw Material Quality Log
1. Raw Material	Surface Defect & Camber Check	Free from laminations, cracks, heavy pitting, surface rust	Max camber/sweep < 1 mm per 1000 mm length	Visual Inspection / Straight Edge Rule	Visual Inspection Report
2. Fabrication	Structural Member Cutting Lengths	ISMC Channels (6000, 2000 mm) & Angle Truss Sections	± 1.0 mm on overall length	Calibrated Steel Rule / Tape	Cutting Dimensional Log
2. Fabrication	Bevel Angles & Mitre Joint Prep	Diagonal truss angles (ISA 100x100x6 / ISA 50x50x4)	± 0.5° bevel angle accuracy	Bevel Gauge / Digital Angle Finder	Fit-up Inspection Log
2. Fabrication	Outer Welding Plate Pitch Spacing	270x100x8 mm Plates with 500.00 mm Gap Spacing	± 1.0 mm Pitch spacing tolerance	Steel Tape / Vernier Caliper	Dimensional Check Report
3. Machining	Hole Diameter Verification	Ø 14.00 mm All Holes across outer plates & main frame	+0.50 mm / -0.50 mm (13.50 mm to 14.50 mm)	Go / No-Go Plug Gauge / Vernier	Hole Diameter QC Log
3. Machining	Geometric True Position Tolerance	True Position: ⌖ Ø 0.5 mm @ MMC with respect to Datum A	Max 0.5 mm deviation from true axis	Drill Jig Check / Layout Coordinate Gauge	Machining Layout Report
3. Machining	Edge Distance, Pitch & Deburring	275 mm / 500 mm centerline pitch. Zero sharp burrs	± 0.5 mm pitch. 100% smooth chamfer	Height Gauge / Visual Inspection	Deburring Clearance Log
4. Fit-up	Door Leaf Overall Frame Dimensions	Overall Height 14,000.00 mm x Width 4,800.00 mm 14,000.00 mm x Width 4,115.00 mm	± 2.0 mm on total height & width	Laser Distance Meter / Steel Rule	Assembly Dimensional Log
4. Fit-up	Frame Squareness & Diagonal Matching	Corner-to-corner diagonal distance measurement	Max Diagonal Difference ≤ 3.0 mm	Cross Steel Tape Check	Squareness Check Sheet

4. Fit-up	Wicket Access Door Special Requirement	Wicket Door Opening (Leaf 02: 1324x2060 mm frame)	STRICT: Do NOT weld in shop; cut & weld on side profile using CO2 welding	Process Quality Verification Audit	Special Constraint Clearance
5. Welding	CO2 Welding Gas & Wire Consumables	CO2 Gas (100% pure) + ER70S-6 Wire (AWS A5.18). SMAW/MMAW prohibited	Gas purity ≥ 99.8%, Wire dia 0.8 / 1.2 mm	Gas Purity Certificate & Wire MTC	CO2 Welding Consumable Approval
5. Welding	Outer Connection Plate CO2 Welds	270x100x8 mm plates welded on all outer edges via CO2 MIG	Continuous fillet weld, throat min 6.0 mm	Weld Gauge / Visual Inspection	Weld Inspection Report
5. Welding	Motor Mounting Bracket CO2 Welds	275x200x16 mm Heavy Bracket Plate (Detail C/D) via CO2 MIG	Full penetration / continuous 10 mm fillet weld	100% Dye Penetrant (DP) / NDT Check	NDT DP Testing Certificate
5. Welding	CO2 Weld Defects & Spatter Clearance	Zero cracks, zero porosity, no spatter, zero undercut	AWS D1.1 Visual Acceptance Criteria	100% Visual Inspection (VT) & NDT	Visual Weld Clearance
6. Coating	Surface Preparation (Shot/Grit Blasting)	Sa 2.5 Near White Metal Blast Cleaning / Wire Brushing	Surface profile depth: 40 - 60 microns	Surface Profile Gauge / Roughness Comparator	Surface Prep Inspection Log
6. Coating	Primer & Finish PU Paint Application	2 Coats Zinc Chromate Epoxy Primer + 2 Coats PU Topcoat	Primer DFT ≥ 50µ, Topcoat ≥ 80µ (Total DFT ≥ 130µ)	Magnetic DFT Gauge (Elcometer)	Paint DFT Inspection Report
7. Dispatch	Identification, Stenciling & Hard Stamping	DWG NO NTPL-0007-PM-001, Project NTPL-0007, Hyderabad Site	100% section hard-stamping & match marks	Visual Verification	Dispatch Clearance Certificate

4. Mandatory Quality Control Standards & Guidelines

- **Material Authenticity & Approved Brands:** All raw structural steel must belong to Grade YST 250 MPa and be manufactured solely by TATA Steel, SAIL, VIZAG (RINL), or JINDAL (JSW Steel). Every batch must be accompanied by original Mill Test Certificates (MTC) stating chemical composition and mechanical properties (Yield ≥ 250 MPa, Tensile 410-540 MPa).
- **Exclusive CO2 Welding Process Requirement:** CO2 Welding (MIG/MAG process with 100% CO2 shielding gas and ER70S-6 wire as per AWS A5.18) is strictly MANDATORY for all shop and site structural joints. Conventional stick electrode welding (SMAW/MMAW) is completely prohibited to ensure high penetration, clean weld beads, and minimal thermal distortion.
- **Wicket Access Door Strict Instruction (Leaf 02):** DO NOT WELD IN SHOP CUT AS PER DIMENSIONS & WELD ON THE SIDE WICKET DOOR TOP & BOTTOM PROFILE. This mandatory instruction prevents structural distortion and misalignment during site assembly in Hyderabad.
- **Datum A True Position & Hole Drilling:** All Ø 14.00 mm holes (+0.50 / -0.50 mm tolerance) must conform to True Position ⌖ Ø 0.5 mm @ MMC referenced to Datum A. Hardened steel drill jigs are mandatory across the 14-meter height.
- **Distortion Control & Welding Sequence:** Due to the extreme panel height of 14,000 mm, staggered and balanced CO2 welding sequences must be strictly implemented for ISMC 125x65 mm channels and ISA 100x100x6 mm angles to eliminate thermal warping, sweep, or camber.
- **Motor Bracket Plate Heavy Weld Inspection:** The 275 x 200 x 16 mm motor bracket mounting plate (Detail C / Detail D) carries high dynamic loads. All perimeter CO2 fillet/butt welds must undergo 100% Dye Penetrant Testing (DPT) as per AWS D1.1.
- **Coating Integrity & Environmental Protection:** Surface preparation must achieve Sa 2.5 cleanliness with a 40-60 micron profile depth. Total Dry Film Thickness (DFT) must exceed 130 microns (50µ Epoxy Primer + 80µ Polyurethane Finish Coat).